Roll No.: .....

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: M.Tech | Semester: III

Name of the Course: Intrusion Detection Systems | Course Code: MCS-324

Time: 3 Hours | **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**

ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are twenty.

iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain the fundamental concepts of intrusion detection and prevention systems (IDS/IPS). How do they differ in functionality and deployment?	CO1
(b)	Discuss different types of IPS systems. Along with the working of Intrusion prevention system.	CO1
(c)	Provide different types of data collected for IDS and IPS along with challenges involved in data collection.	CO2
Q2		
(a)	The effectiveness of an Intrusion Detection System (IDS) largely depends on where it is deployed within a network. Analyze different IDS placement strategies.	CO2
(b)	Explain how a Genetic Algorithm (GA) can be applied to optimize or design an Intrusion Detection System (IDS). Discuss each operation of the GA with a suitable example showing how attack detection rules can evolve over successive generations.	CO3 CO3
(c)	Analyze the role of machine learning techniques in improving anomaly detection accuracy.	
Q3		
(a)	Explain the three-tier architecture of Intrusion Detection and Prevention Systems (IDS/IPS). Discuss the roles and interactions of different components in ensuring network security.	CO4 CO4 CO4
(b)	Analyze how laws, standards, and security organizations influence the management and deployment of IDS/IPS systems.	
(c)	Discuss major business issues that organizations face while implementing and maintaining security infrastructure.	
Q4		
(a)	Explain the information flow within IDS/IPS systems. Discuss how each stage contributes to accurate intrusion detection.	CO5 CO5 CO5
(b)	Describe the internal mechanisms used by IDS/IPS for exploit detection and malicious code detection.	
(c)	Intrusion Detection and Prevention Systems (IDS/IPS) themselves can become targets of sophisticated attacks. Analyze the key challenges in defending IDS/IPS systems	

	against such attacks along with defense strategies.	
Q5		
(a)	Analyze the challenges involved in configuring event filters in large enterprise networks. How can improper filter design affect the overall performance and security posture of the organization?	CO6 CO6
(b)	Evaluate the importance of continuous signature updates and version upgrades in maintaining effective intrusion detection. Provide examples of risks associated with outdated signature databases.	CO6
(c)	Discuss the role of data correlation and incident response in effective IDS management. How do well-defined policies and procedures ensure timely and coordinated security operations in an organization?	